

Moyamoya Disease: A Review of the Disease and Anesthetic Management

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Abstract: Moyamoya disease is a rare chronic cerebrovascular disease seen both in children and adults. It has a progressive course, but may have a variable clinical presentation. The disease causes ischemic stroke, intracranial hemorrhage, headache, seizures, and transient ischemia attack in children and in adults. Although the pathogenesis of the disease remains unknown, research suggests a genetic predisposition. There are also undefined systemic processes involved in this vasculopathy. Better noninvasive diagnostic techniques for diagnosis of the Moyamoya disease have been developed, but medical treatment can still be challenging. However, various surgical revascularization procedures have shown to provide symptomatic benefit in a majority of these patients. In addition, the anesthetic management of these patients has evolved over the years with an increased understanding of the disease. These have specifically resulted from the identification of risk factors for perioperative complications and outcomes related to the use of anesthetic agents. Finally, research in the last 3 decades has led to the recognition of the importance of pain control, the increased use of regional anesthesia, and better monitoring techniques in providing high quality and safe patient care to patients with Moyamoya disease. This article will provide a comprehensive review of the disease and its anesthetic management.

Key Words: Moyamoya disease, anesthetic management, cerebrovascular disease, ischemic stroke, transient ischemic attack

(*J Neurosurg Anesthesiol* 2011;23:100–109)

Moyamoya disease (MMD) was first described by Takeuchi and Shimizu¹ in 1957 and later by Nomura² in 1961. It is characterized by angiographic evidence of progressive stenosis or occlusion of terminal portions of the internal carotid arteries and the proximal portion of the anterior and middle cerebral arteries. The

posterior cerebral arteries may also be involved. The formation of reticular anastomosis and collaterals between the internal and external carotid arteries gives the image of a “puff of smoke” on angiogram.³

EPIDEMIOLOGY

MMD has definite ethnic and familial predisposition. The incidence of MMD is highest in Japan with an incidence of 0.54, and crude prevalence of 6.03, per 100,000 population.⁴ It is also higher in Korea and other Asian countries than in the rest of the world. It has a female predominance, with a female:male ratio of 1.8:1.⁴ In Japan, the prevalence in males shows peaks in the 10 to 14, 35 to 39, and 55 to 59-year-old age groups whereas in women the prevalence is higher in the 20 to 24 and 50 to 54-year-old age groups.⁴

The review of epidemiological data of MMD in the United States showed an incidence of 0.086/100,000 persons. The ethnicity-specific incidence rate ratios were 4.6 for Asian Americans, 2.2 for African Americans, and 0.5 for Hispanics compared with Caucasians.⁵ Survey of MMD in Hawaii reported higher prevalence among people of Japanese descent compared with Caucasians and also higher prevalence among non-Japanese Asians and Pacific Islanders compared with Whites.⁶ Patients with MMD in North America are more likely to be young adults whereas in Asia they are more likely to be children.⁷ In the United States the affected patients usually present with ischemic symptoms, although the rate of hemorrhage is higher in adults than children (20% vs. 2.8%, respectively).^{7,8}

ETIOLOGY AND PATHOLOGY

MMD is probably a genetically inherited disease, as a familial occurrence is reported in 9% to 12% of cases.⁴ The incidence of the disease in both monozygotic twins is 80% and the incidence rate in offspring of the affected people is 34 times higher than the incidence in the general population.^{9–11} Genetic studies conducted on Japanese families with MMD suggest linkage to multiple chromosomal regions: 3p24.2-p26, 6q25, 8q23, 12p12, and 17q25.^{9,12–15} The linkage to different loci suggests the disease is possibly polygenic or may represent locus heterogeneity.⁹ MMD is inherited most likely in an autosomal dominant manner with low penetrance or in a polygenic mode.^{11,12} Genome-wide parametric analysis

Received for publication April 6, 2010; accepted August 19, 2010.
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Financial Support: No financials supports, no commercial affiliations or any disclosures to report.
Conflict of Interest: None identified.
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located the gene locus for autosomal dominant MMD on chromosome 17q25.3.¹²

The pathogenesis of the disease remains unknown. The histopathology of the involved vessels in the patients with MMD shows eccentric intimal thickening by fibrous tissue with occasional lipid proliferation.¹⁶ Vessel occlusion occurs from hyperplasia of smooth muscle cells and luminal thrombosis.⁸ Inflammatory or atheromatous changes are absent, but rarely hypoplasia of the vessel with absent muscularis is seen. Some associated findings are seen in the patients with MMD although their role remains unclear. The findings suggest undefined systemic process (possibly nongenetic) may play a role in the vasculopathy in some patients with MMD¹⁷; basic fibroblast growth factor is increased in the cerebrospinal fluid (CSF) and arteries¹⁸ and there are increased levels of soluble vascular-cell adhesion molecule type 1, intercellular adhesion molecule type 1 and E-selectin in CSF.¹⁹ Increased levels of soluble endothelial adhesion molecules have been showed in CSF suggesting that patients with MMD have ongoing central nervous system inflammation with slight impairment of the blood brain barrier.¹⁹ Increased endothelial progenitor cell numbers are seen in MMD, which may have a role in angiogenesis and arteriogenesis.²⁰ Some investigators suggest that the formation of collateral vessels is induced by the occurrence of arterial stenosis in the hypoxic regions of the brain.²¹ Various other mechanisms implicated in the development of stenosis and occlusion of vessels in MMD include: caspase-dependent apoptosis, direct vessel injury, and immunological targeting.^{22–24} The role of radiation and certain types of infections have been implicated in the pathogenesis of MMD in some patients.^{13,17,25} The association of MMD with specific medical conditions is

TABLE 1. Medical Disorders Associated With Moyamoya Disease^{10–14}

Congenital syndromes	Apert syndrome Downs syndrome Hirschsprung disease Marfan syndrome Neurofibromatosis type 1 Turner syndrome
Immunologic disorders	Graves disease
Hematologic disorders	Aplastic anemia Fanconi anemia Sickle cell anemia Thalassemia
Neoplasia	Wilms tumor Craniopharyngioma parasellar tumors
Infectious	Tuberculous meningitis Leptospirosis
Vascular disorders	Atherosclerotic disease Cardiomyopathy Coarctation of aorta Hypertension
Others	Radiation injury Pulmonary sarcoidosis Livedo reticularis Nephrotic syndrome

TABLE 2. Clinical Presentation of Moyamoya Disease^{17,31–33}

Symptoms	
Frequent	Ischemic stroke Transient ischemic attack Headache
Occasional	Seizure Choreiform movements Intracranial hemorrhage (common in adults uncommon in children)
Rare	Cognitive decline Visual impairment

unclear, but could be related to the genetic basis of MMD (Table 1).^{25–29}

CLINICAL PRESENTATIONS

MMD is usually symptomatic, although asymptomatic cases have been reported.³⁰ It can cause a range of symptoms including ischemic stroke, transient ischemic attack (TIA), intracranial hemorrhage, seizures, and headaches (Table 2).^{17,31–33} Soriano et al.³¹ noted that symptoms in children may start in infancy and may cause rapid deterioration in the child's neurological function. A 2009 study described the initial presentation of 96 children and 233 adults and reported that presentation differs with age and the single most common presentation is ischemic stroke in both children and adults.³² In the affected children, stroke, TIA, and headache were common presentations whereas intracranial hemorrhage was rare. Headaches and repeated TIA may occur during crying, exercise, coughing, or straining.^{4,16,17,33} These children may also present with slow cognitive decline, dizziness, seizures, visual impairment, involuntary movements, hemiparesis, monoparesis, sensory impairment, or cerebral infarction.^{17,33} In adults, common presentation includes stroke, TIA, and headache.³² Adult MMD patients in the United States have a predominance of ischemic stroke.^{32,33} Other common presentation findings in adults are seizures and intracranial hemorrhage, including atraumatic acute subdural hematoma resulting from spontaneous rupture of transdural collateral vessels.^{17,30} Patients may also present with growth hormone deficiency, possibly resulting from chronic cerebrovascular insufficiency.^{33,34}

MMD progresses eventually to complete occlusion of the internal carotid artery.³ Approximately two thirds of patients with MMD have a symptomatic progression over a 5-year period.³⁵ Research indicates that the occlusive arterial change in MMD is an active process.^{35,36} In addition, the disease progression occurs unilaterally or bilaterally in both symptomatic and asymptomatic patients and may involve both anterior and posterior circulations.^{36,37}

DIAGNOSIS

The workup for suspected MMD starts with a head computed tomography (CT) scan based on clinical

TABLE 3. Angiographic Staging of Moyamoya

Grade I	Carotid stenosis without collateral vessels
Grade II	Basal collateral vessels seen
Grade III	Prominent basal collateral vessels
Grade IV	Stenotic or occluded circle of Willis and posterior cerebral arteries
Grade V	Extra cranial collateral network
Grade VI	Total carotid occlusion

Adapted with permission from *Arch Neurol*. 1969;20:288–299.

findings or if intracranial hemorrhage is suspected. Patients may be evaluated with magnetic resonance imaging (MRI)/magnetic resonance angiography (MRA) that is able to show acute infarcts, chronic infarcts, and a cortical ischemic pattern.³⁸ The diagnostic MRI findings suggestive of MMD are diminished flow voids in the internal carotid and anterior and middle cerebral arteries with prominent collateral flow voids in the basal ganglia and thalamus. MRA is now more popular as a primary diagnostic technique for diagnosis of MMD because it is noninvasive and provides good imaging results whereas cerebral angiography remains the gold standard to confirm the diagnosis of MMD.^{39–41} The risk of neurologic complications in cerebral angiography in children with MMD is low and statistically not different from the risk in children with other cerebrovascular disorders.⁹ On the basis of angiographic evidence of severity of the disease, MMD is classified into 6 stages that correlate with the disease progression (Table 3).³ Other noninvasive cerebral blood flow studies that are not often used but possibly helpful in diagnosis include: transcranial Doppler ultrasonography, xenon-enhanced computed tomography, positron emission tomography (PET), and single-photon emission computed tomography with acetazolamide challenge.¹⁷

Electroencephalogram (EEG) does not diagnose MMD, but shows interesting changes in these patients. For example, high-amplitude slow waves are seen on EEG in a normal child with hyperventilation, which disappear when the child resumes normal breathing, but in a child with MMD the slow waves then reappear. This reappearance of slow waves in the absence of hyperventilation is called “re-build up phenomenon” and is characteristic of MMD.^{7,16,42} Guidelines for the diagnosis of MMD have been proposed by the Ministry of Health and Welfare, Japan.⁴¹

MEDICAL TREATMENT

Medical treatment does not stop the progression of MMD. A 2006 study of Moyamoya patients who were treated conservatively, found that the 5-year risk of recurrent ipsilateral stroke was 65% in symptomatic patients with unilateral involvement whereas in symptomatic patients with bilateral involvement the risk increases to 85%.⁷ The medications used in medical management are: antiplatelet agents (usually aspirin or ticlopidine) and vasodilators such as calcium channel blockers (nimodipine, nicardipine) and pentoxifylline.^{33,43,44} Anticonvul-

TABLE 4. Surgical Procedures for Moyamoya Disease^{8,15,17}

Direct Bypass Techniques: most commonly done with a superficial temporal artery (STA) to middle cerebral artery (MCA) bypass. This is procedure provides immediate increase in blood flow to ischemic brain, usually done in adults and is technically difficult to conduct in children
Encephaloduroarteriosynangiosis (EDAS): preferred indirect revascularization procedure in which the STA is mobilized and then sutured to the edges of the opened dura. This is usually done in children but in adults is combined with STA-MCA bypass technique
Pial synangiosis: a modification of EDAS in which the surface of the dura is exposed and the pial surface is increased by opening it into multiple flaps to increase collateral vessel development from the dural vessels. A branch of the STA is sutured directly into the pial surface. Results from study indicate majority of pediatric patients with MMD undergoing pial synangiosis stop having strokes and TIAs
Encephalomyosynangiosis (EMS): an indirect revascularization procedure in which temporalis muscle is attached to the surface of the brain to promote collateral vessel formation
Other indirect anastomatic procedures: drilling of burr holes without vessel synangiosis
Encephaloduroarteriomyosynangiosis (EDAMS): combination of EDAS and EMS. Dural flaps created are folded into the dural/epiarachnoid space with middle meningeal artery promoting angiogenesis. Neovascularization reported in 50–90% cases in different studies. Incidence of postoperative strokes is reduced when combined with STA-MCA bypass
Other combination procedures: encephalomyoarteriosynangiosis (EMAS), a combination of EDAS and EMS; encephaloduroarteriomyosynangiosis
Other surgical procedures: craniotomy with dural inversion, cervical sympathectomy, and omental transposition

sants are used in those having seizures. Aspirin is prescribed lifelong at some centers; the daily dosage is 81 mg in children and 325 mg in adults.⁸

SURGICAL TREATMENT

Patients with recurrent or progressive cerebral ischemic events and reduced cerebral perfusion are candidates for surgery. The goal of revascularization procedures is to increase collateral blood flow by promoting neoangiogenesis and inducing collateral vessel formation.^{17,45} There are many different operative techniques, but the most common are either direct or indirect type of bypass procedures (Table 4).^{8,17,30,32} The type of revascularization procedure carried out depends on the patient's age, symptoms, comorbidities, and the surgeon's expertise.⁴⁵ The more symptomatic side in a patient with bilateral MMD and the nondominant hemisphere in an asymptomatic patient is revascularized first.³²

The common direct revascularization procedure is superficial temporal artery (STA) to middle cerebral artery (MCA) bypass.^{17,45} The patency rate and the clinical outcome for this procedure are good.⁴⁵ With instant improvement of focal cerebral perfusion, it decreases the recurrence of cerebral ischemic episodes.^{17,46} However, this procedure is technically difficult in children because of the small size of the arteries.^{17,45} Complications include ischemic injury owing to arterial occlusion, wide blood pressure changes, seizures, and

symptomatic hyperperfusion owing to increased cerebral blood flow (CBF).^{45,47}

Many indirect bypass procedures have been developed. Encephaloduroarteriosynangiosis (EDAS) is carried out by dissecting the STA free and suturing the patent artery to the cut edges of the opened dura. This promotes angiogenesis and reduces recurrent ischemic episodes, but it takes months to develop adequate collateral circulation and symptomatic relief.^{17,45} Also, the direct STA-MCA procedure cannot be carried out once the STA has been used in EDAS.⁴⁵ Encephalomyosynangiosis (EMS) is an indirect bypass procedure in which the temporalis muscle is dissected and placed on the surface of the brain to promote collateral vessel formation.¹⁷ Encephalomyoarteriosynangiosis (EMAS) is a combination of EDAS and EMS, but the efficacy of this procedure is not well established.^{8,17,45}

Pial synangiosis is a modification of the EDAS procedure. In this procedure, multiple dural flaps are made and the arachnoid mater is removed in the area of the dural opening. The posterior branch of the STA is dissected and attached to the pial surface to enhance collateral vessel formation.^{17,45} Data from a study of 143-MMD patients who underwent pial synangiosis procedure reported only 3.2% had strokes after 1 year compared with 67% preoperatively.^{17,35}

Encephaloduroarteriomyosynangiosis (EDAMS) is another modification that combines the placement of temporalis muscle, a segment of STA or middle meningeal artery and a section of galeal flap over the involved cerebrum to promote angiogenesis.⁴⁵ Omental transposition is carried out by tunneling vascular omentum subcutaneously from the abdomen or by transplanting an excised portion with surgical anastomosis between the vascular pedicle and STA.^{45,48} Patients undergo both craniotomy and laparotomy for this procedure and the patency rate for the graft has been reported as high as 70%.⁴⁵ Another technique is the creation of burr holes to promote cerebral neorevascularization. In a study of patients who underwent burr hole procedures, angiograms done postoperatively showed that neovascularization occurred in 41 of 43 burr holes.⁴⁹ This procedure can be carried out under local anesthesia and the incidence of perioperative complications is less than for invasive surgical procedures.⁴⁵

Children are usually treated by indirect revascularization because direct bypass is technically difficult and the success of angiogenesis is greater in children than adults.⁴⁵ Adults are treated by direct revascularization or by combination of direct and indirect techniques as the progression of MMD is not always corrected by STA-MCA bypass. The success of surgical revascularization has been evaluated by various studies. In fact, a meta-analysis concluded that revascularization surgery with direct, indirect, and combined techniques provided symptomatic relief in 87% patients (1003 out of 1156).⁵⁰ Long-term follow-up studies of patients after revascularization surgery have concluded that beyond the safety and symptomatic benefits of surgery, the patients also had a 96% probability of remaining stroke free over the next

5 years.^{35,51} A recent study showed the effectiveness of revascularization procedures in MMD, and reported that the cumulative 5-year risk of stroke or hemorrhage after revascularization was 5.6% compared with 65% in symptomatic patients without surgery.³²

ANESTHETIC MANAGEMENT

There is no evidence associating the use of a particular anesthesia technique or drug with better outcome after induction or maintenance of anesthesia in MMD patients.³¹ A variety of techniques and medications are used, each with advantages and disadvantages. The goal for anesthesia is to maintain the balance between the oxygen supply and demand. In MMD it becomes more important to avoid neurologic morbidity because of the underlying pathophysiology. CBF should be maintained by avoiding hypotension and maintaining normocarbida while an increase in cerebral oxygen consumption (CMRO₂) associated with laryngoscopy, tracheal intubation, and surgical events can be minimized by the use of appropriate depth of anesthesia.^{17,31}

ANESTHETIC EVALUATION

A thorough and detailed preoperative anesthetic assessment is vital. Many medical conditions have been reported in association with MMD, which may impact anesthetic management.^{25–29} History of frequent preoperative TIA indicates compromised cerebral blood supply and is a significant risk factor for perioperative complications.⁵² Many of these patients have neurologic deficits including segmental motor deficits, epilepsy, and mental retardation as the result of chronic ischemia.^{17,30,53} Review of a MMD patient's chronic medications is essential. Antiseizure medications and calcium channel blockers should be continued until the day of surgery. Low-dose aspirin is continued at some centers until the day of surgery and at some centers it is stopped 7 to 10 days before the surgery and replaced with low-molecular weight heparin (LMWH). LMWH 0.5 mg per kg twice a day subcutaneously is administered in some centers for neurologically unstable patients who need reversible anticoagulation before the surgery and angiography. Aspirin is restarted after the surgery on the first postoperative day.¹⁷

It has been recommended that MMD patients be admitted on the evening before surgery for aggressive intravenous hydration.¹⁷ For outpatients, if the surgery is delayed for some reason the patient may be given clear fluids as per NPO guidelines or may receive intravenous fluids to avoid dehydration that may result in decreased CBF.

PREMEDICATION AND INDUCTION

Premedication for anxiolysis may be beneficial in children with MMD as crying causes hyperventilation resulting in hypocapnia with possible cerebral vasoconstriction, decreased CBF, and cerebral ischemia.^{17,54–56} There are no evidence-based data suggesting any advantage

or better outcome with any particular premedication although midazolam is most commonly used.⁵⁵

Induction may be achieved with inhalation or intravenous techniques.⁵⁷ Inhalation induction in children is achieved with sevoflurane while avoiding hyperventilation (that will cause hypocapnia). Inhalation induction should be titrated to avoid hypotension and impairment of cerebral autoregulation. Intravenous (IV) induction is also routinely done after the application of a topical anesthetic cream at the venous access site in children.⁵⁵ Intravenous induction agents: thiopental, propofol, etomidate with possible exception of ketamine may be used.^{55,57,58} The choice of a nondepolarizing muscle relaxant should be less likely to cause histamine release and/or any hemodynamic changes.⁵⁵ Patients should be intubated once an adequate depth of anesthesia is achieved and muscle paralysis is confirmed (if required). The administration of intravenous opioids and lidocaine may help to attenuate the laryngoscopy and tracheal intubation response.⁵⁸ Hypoxemia, hypercarbia, coughing, and bucking must be avoided with laryngoscopy and endotracheal intubation. Induction should be smooth to avoid increases in intracranial pressure (ICP) and decreases in cerebral perfusion pressure (CPP).

MONITORING

The monitoring for most cases (with minimal hemodynamic instability and fluid shifts) includes the use of standard ASA monitors.⁵⁵ A patient undergoing a major surgical procedure involving hemodynamic instability, significant blood loss, or comorbidity may also require invasive monitoring with arterial blood pressure, central venous pressure, and urine output. Cerebral ischemia is commonly monitored by EEG, although near infrared spectroscopy (NIRS), somatosensory-evoked potentials, and transcranial doppler monitoring are used in a few centers.^{17,55,59} However, there is no evidence that the use of NIRS or transcranial doppler monitoring leads to the earlier detection of cerebral ischemia or favorably impacts the outcome.

MAINTENANCE OF ANESTHESIA

Anesthesia may be maintained with a volatile agent and opioids or with total intravenous anesthesia (TIVA) using propofol and opioids with or without a low dose of volatile agent. The use of nitrous oxide is controversial and there exists no clear consensus regarding its use in patients with MMD. It has been used in patients undergoing revascularization procedures and during cardiopulmonary bypass in patients with MMD.^{57,58,60}

Investigators analyzed the perioperative course of 368 cases of revascularization surgery in which nitrous oxide was used and concluded that the severity of disease and surgical procedures were major determinants of neurological deterioration, rather than anesthetic management that involved the use of nitrous oxide.⁵⁷

Most of the currently available volatile agents have been used in patients with MMD.^{57,58} All volatile anes-

thetics cause cerebral vasodilatation and an increase in CBF with reduction in CMRO₂. Both isoflurane and sevoflurane are commonly used, although clear evidence of which is more advantageous is still lacking. An early report from 1999 compared the effect of sevoflurane and isoflurane on dynamic cerebral autoregulation using transcranial doppler at 1.5 minimum alveolar concentration (MAC). The conclusion was that dynamic autoregulation is better preserved during sevoflurane than isoflurane anesthesia.⁶¹ More recently, the effect of inhalation anesthesia and TIVA on cerebral circulation in MMD has been studied.⁶² This study measured regional cortical blood flow, jugular bulb oxygen saturation, and frontal regional oxygen saturation by NIRS in patients undergoing revascularization procedures. The investigators concluded that regional cortical blood flow (rCoBF) levels may be decreased with inhalation anesthesia and may provoke intracerebral steal; this was not seen with TIVA.

Some studies have shown improved outcome with TIVA using propofol, whereas other studies have shown no significant difference as compared with isoflurane.^{63,64} Thus, there is reportedly no significant difference in postoperative complications between the two. Similarly, the incidence of early postoperative complications under inhaled anesthesia versus TIVA in patients with MMD undergoing revascularization procedures has been reviewed and the authors found no significant differences in the postoperative complications of the 2 groups.⁶³ However, a recent study compared the effect of sevoflurane and propofol on ICP and rCoBF using a laser doppler flowmeter probe during revascularization procedures in patients with MMD and concluded that rCoBF was significantly higher and ICP lower with propofol.⁶⁴

VENTILATION

Carbon dioxide is a potent modulator of cerebrovascular tone and influences the CBF with changes in ventilation and can be a major factor in determining neurologic complications.³¹ In Moyamoya patients hypercapnia dilates the normal cerebral vessels, whereas the diseased vessels show minimal response resulting in less blood flow to areas supplied by diseased vessels.⁵² The effect of hypercapnia perioperatively on cortical blood flow in patients with MMD using a laser-doppler method has been studied.⁶⁵ The study concluded that there was a decrease in regional cortical blood flow with hypercapnia so that the normal cortical blood flow response to hypercapnia was impaired during surgery. Evidence from a study of cerebrovascular response to hypercapnia using PET concluded that cerebrovascular response to hypercapnia was severely decreased over the cerebral cortices in patients with MMD.⁶⁶ Similar findings have been reported in children with MMD.⁶⁷ Hypocapnia induced by hyperventilation causes cerebral vasoconstriction and the areas of brain supplied by diseased vessels are more at risk of cerebral ischemia.⁵² Hyperventilation induced by crying or exercise has been seen to trigger TIA in children with MMD.⁵⁴⁻⁵⁶ The regional blood flow in children with

MMD during rest and hyperventilation has been studied using 133-Xe inhalation methods.⁶⁸ During hyperventilation decreased regional cerebral blood flow was observed in all patients when the PaCO₂ decreased below 29 mm Hg; hence the goal in patients is to maintain normocarbica with controlled ventilation, as reported earlier by Chiu et al.³³

BLOOD PRESSURE AND INTRAVASCULAR VOLUME

It is important perioperatively to maintain the blood pressure at or above the preoperative baseline.³³ A fall in mean arterial blood pressure will result in a decrease in CBF and CPP with resultant cerebral ischemia and infarction.^{52,62} Reduction in CBF is better tolerated in adults than children who have higher CMRO₂.³¹ Autoregulatory response to hypotension is substantially diminished in children compared with adults.⁶⁷ Children with MMD have greater higher oxygen extraction ratio than adults, explaining the higher frequency of ischemic symptoms.⁶⁶ Children with MMD are prone to develop neurologic deficits during hypotension because of the mismatching of their CBF and CMRO₂.³¹ Patients can become hypotensive perioperatively because of the anti-hypertensive medications given preoperatively, dehydration, inadequate volume replacement during surgery, blood loss, and the effect of anesthetic agents. Takagi et al⁶⁹ reported a case of MMD with renovascular hypertension and hemodynamic cerebral infarctions owing to reduced blood pressure brought on by anti-hypertensive therapy preoperatively. There have been recommendations that patients should be kept in a hypervolemic state to avoid the problems related to hypotension and decreased CPP.^{17,70} Hypotensive anesthetic techniques should be avoided. Hyperosmotic drugs should be avoided as they cause dehydration and hypotension. Dopamine, phenylephrine, and ephedrine have been used to treat hypotension.^{55,58,60} If the ICP is elevated, ventricular drainage may be used instead of administering hyperosmotic drugs.⁷¹ Many of the anesthetic agents and medications have the potential to lower the blood pressure, and therefore must be carefully titrated. Kim and Yoo⁷² described a case of MMD in which off-pump coronary artery bypass grafting (OP-CAB) was done to avoid cardiopulmonary bypass and the risk of intraoperative hypotension.

HEMATOCRIT

A decreased hematocrit owing to anemia or perioperative blood loss places MMD patients at the risk of cerebral ischemia. If severe, anemia should be corrected preoperatively with blood transfusion to improve oxygen delivery. In procedures involving extensive blood loss, it becomes critical to regularly check the hematocrit intraoperatively and transfuse blood as necessary. There is no consensus at what hematocrit the patient with MMD should be transfused. An ideal hematocrit is debatable, but a hematocrit of 30% to

42% has been proposed as adequate.^{73,74} Polycythemia with associated increased viscosity is regarded as a risk factor for cerebral infarction in patients with MMD.⁷⁰ Hemodilution is used to improve cerebral perfusion and avoid cerebral ischemia, but if excessive can itself cause cerebral ischemia by reducing the oxygen-carrying capacity of the blood.

TEMPERATURE

Mild hypothermia is regarded as offering some degree of neuroprotection against cerebral hypoxia and ischemia by reducing the cerebral metabolic rate. However, there have been few studies and no randomized clinical trials conducted. Over 20 years ago, investigators postulated that a drop in body temperature may induce vasospasm.⁷⁵ Later, Wang et al⁶⁰ described a patient with MMD who underwent cardiac surgery and cardiopulmonary bypass for atrial septal defect. Importantly, this patient had no neurologic deficit even when body temperature was lowered to 34°C during the CPB. Wang and colleague's results were consistent with work by Kurokawa et al⁷⁶ who showed that hyperthermia increases cerebral metabolic rate, which causes an increase in oxygen consumption and possible cerebral ischemia. Other experts also recommend avoiding hyperthermia.¹⁷ On the basis of findings for the last 25 years, it can be generally agreed that normal body temperature should be maintained perioperatively.

URINE OUTPUT

Urine output generally reflects the volume status of the patient under anesthesia (with normal cardiovascular and renal function).⁵⁵ Low urine output intraoperatively has been identified as a potential risk factor associated with complications in patients with MMD.⁷⁰ Evidence exists supporting that intravascular volume should be kept in the range of hypervolemia.^{17,70} Results from the study concluded that ischemic complications occurred in a group of patients with lower urine output (2.2 mL/kg/h) compared with no incidence of complications in the group with greater urine output (4.1 mL/kg/h).⁷⁰

SPECIFIC ANESTHETIC CONSIDERATIONS DURING REVASCULARIZATION PROCEDURES

The tenuous cerebral blood supply places patients with MMD at increased risk of perioperative morbidity during revascularization procedures because of the hemodynamic changes that may occur with anesthesia.^{77,78} Michenfelder⁷⁹ recommended using the same anesthetic management techniques as for the carotid artery procedures for revascularization procedures. Other experts have recommended the same anesthetic considerations in revascularization surgery as with a case of arteriovenous malformation.⁸⁰ The general consensus is to maintain CPP at or above the baseline. The autoregulation of CBF to the ischemic cerebral areas is impaired and the CBF is totally pressure dependent.⁷⁹ Thus during revascularization techniques blood pressure is maintained in the

normal-to-high range.⁷⁷⁻⁷⁹ Maintenance of normocapnia is recommended in revascularization procedures.⁷⁷⁻⁷⁹ Investigators also recommend maintaining hematocrit of greater 30% to minimize perioperative ischemic complications.⁷⁸ During STA-MCA bypass neuroprotective measures are used including mild hypothermia and barbiturate induced burst suppression using sodium thiopental.^{77,78,81} STA-MCA anastomosis provides low-flow revascularization because of the relatively small diameter of the recipient artery.⁸² This can result in cerebral hyperperfusion causing transient focal neurological deficit or delayed intracerebral hemorrhage. Adult-onset patients have a significantly higher risk for symptomatic hyperperfusion compared with child-onset patients. The incidence of symptomatic hyperperfusion is higher in adults (38.2%) and lower in children (5.9%) after STA-MCA anastomosis.⁸³

REGIONAL ANESTHESIA IN MMD

Regional anesthesia including central neuroaxial block and peripheral nerve blocks may be used depending on the type of surgical procedure and the patient's comorbidities. These blocks should be attempted in children only when the patient is under anesthesia to reduce anxiety and any discomfort. Regional anesthesia will provide or potentiate both intraoperative and postoperative analgesia and facilitate smooth awakening. It is the patient's comorbidities and hemodynamic status rather than age that can be a limiting factor for regional anesthesia.

Caution should be used with subarachnoid blocks that can cause high spinal or total spinal block with severe hypotension and cerebral ischemia. Convulsions and temporary hemiparesis after spinal anesthesia in a child with MMD have been reported.⁸⁴ The epidural route can be used to provide analgesia in surgical procedures with the advantage that the level of block and degree of hypotension can be controlled. There are case reports of the use of epidural for labor and cesarean section in patients with MMD.⁸⁵⁻⁸⁷ Abouleish et al⁸⁸ reported a case of elective cesarean section of a MMD patient anesthetized by combined spinal and epidural technique. They reported that it allowed better analgesia than epidural anesthesia and more hemodynamic stability than either general or spinal anesthesia alone. The use of skull block (ie, combined supraorbital, supratrochlear, auriculotemporal, and posterior auricular nerve block) during EDAMS surgery in children age 3 to 13 years with MMD has been shown to provide calm awakening, better pain relief, and possibly reduced postoperative morbidity.⁸⁹

POSTOPERATIVE CARE

Patients who are asymptomatic and hemodynamically stable undergoing noninvasive procedures (usually imaging studies) may be discharged on the same day. Patients undergoing revascularization procedures are kept in intensive care unit (ICU) overnight.³² The postoperative neurological course of some patients (who

are hemodynamically or neurologically unstable or with a history of neurological event or significant comorbidities) can be unpredictable as such they should be admitted for observation or may require admission in ICU even if undergoing noninvasive procedures. Patient undergoing other invasive procedures are usually admitted and may need also admission in ICU depending on their surgical procedure, neurologic, and hemodynamic status. If patients develop any neurologic changes they should be immediately evaluated by both the neurologic and radiologic examinations as indicated.

Blood pressure, central venous pressure, urine output, hematocrit and oxygen saturation should be closely monitored in the postoperative period. After revascularization procedures, it is important to avoid both hypertension that may cause bleeding and hypotension that may cause graft thrombosis. It is recommended that mean blood pressure be maintained in the range of 80 to 100 mm Hg in adults.³² Postoperative intravenous fluids at one and one-half times the normal maintenance rate and starting aspirin on the first postoperative day have been recommended to address this issue.¹⁷

PAIN CONTROL

Pain is associated with neuroendocrine responses, causing an increase in cerebral metabolism that is undesirable in patients with MMD. Postoperative cerebral infarction may be associated with inadequate pain control.⁷⁴ Adequate analgesia can be achieved with a proper pain management strategy depending on the surgical procedure, age of patient, and comorbid conditions. Perioperative analgesia may be achieved with opioids that can be given by titrating intravenous doses, by continuous infusion, or regional anesthesia or a combination of both intravenous and regional techniques.

Good postoperative analgesia may reduce the risk of cerebral ischemia or infarction associated with crying.^{54-56,85} Hypercarbia is a concern with opioid use and may result in less blood flow to areas supplied by diseased vessels.⁵² Intramuscular injections should be limited as a route of administration because they are often painful stimuli that can increase anxiety and pain in some pediatric patients. Patients must be made comfortable by providing additional analgesia or sedation as needed for dressing changes and removal of drains and catheters. For more common or simple procedures, the use of resorbable subcuticular sutures with steristrips for wound closure followed by soft paraffin gauze over the wounds that are then covered by a dressing has been recommended.⁵⁶ These researchers advocated the use of local anesthetics to provide analgesia for placing intravenous lines and keeping central venous lines in place for a week after surgery.

RISK FACTORS, PERIOPERATIVE COMPLICATIONS, AND OUTCOME

The perioperative complications in MMD patients are influenced by the type of procedure (whether

diagnostic or invasive), type of revascularization procedure, severity of disease, and nonsurgical hemodynamic risk factors.^{33,52,57,70} Patients with such risk factors remain at greater risk for ischemic neurologic events during the perioperative period and may develop stroke, TIA, intracranial hemorrhage, and infection.^{8,90} A study by Iwama et al⁵² found a history of TIA to be a significant risk factor for perioperative complications and patients who were hypercapnic had increased perioperative ischemic complications. They concluded that hypotension and hypocapnia did not independently affect the occurrence of postoperative ischemic complications because their duration was controlled. A separate study of the perioperative course in patients who underwent revascularization surgery found a history of frequent TIA to be a risk factor for postoperative ischemic complications in patients with MMD.⁵⁷

Sato et al⁷⁰ assessed perioperative ischemic complications in patients with MMD and identified 3 risk factors: (1) decreased intraoperative urinary output, (2) decrease in the hematocrit, and (3) preoperative low-density areas (LDA) seen on preoperative CT scan. They further noted that intraoperative hypercapnia, hypotension, and reduced circulating blood volume were also associated with complications.

Evidence suggests that sparing vital collateral vessels and minimum brain retraction during surgery are essential to avoid perioperative complications.⁹¹ Postoperative ischemic complications are fewer in patients who have combined procedures (STA-MCA bypass with EDAMS) compared with indirect bypass.⁹² Moreover, patients who underwent indirect revascularization had a greater risk of neurologic deterioration than those who had direct anastomatic procedures.⁵⁷ Guzman et al³² carried out 450 revascularization procedures in 329 patients with a surgical morbidity of 3.5% and a mortality of 0.7% per treated hemisphere. In patients who had initially presented with TIA, the majority of them (91.8%) were free of TIA at 1 year or later. The risk for stroke is 4% for revascularization surgery in the first month, which then decreases and patients have a 96% probability of remaining stroke free for 5 years.^{50,51}

Because of the relatively small number patients with MMD, there are very few randomized trials to evaluate long-term benefits of surgical revascularization procedures. Park et al⁹³ evaluated the surgical outcomes of MMD that were measured by activity of daily living (ADL) and prognostic factors. They found that the improvement in ADLs was not statistically different (59.1%) in patients who had surgical revascularization procedures compared with those treated conservatively (57.5%). However, the incidence of recurrent infarction in the patients who had surgery was lower compared with those treated conservatively (5.9% vs. 26.3%).

CONCLUSIONS

Moyamoya remains a disease with unknown etiology that results in a challenge in determining medical

treatment. Early diagnosis and effective management are necessary before any ischemic event and neurologic impairment occurs. Surgery remains the only viable option to decrease further ischemic insult and neurologic deterioration.⁸ Surgical revascularization procedures have been shown to provide symptomatic benefit in 87% of patients.⁵⁰ As a result, MMD patients are likely to require anesthesia for multiple diagnostic and surgical procedures. Thus, safe and effective anesthetic care of these patients involves understanding the pathophysiology of the disease and its anesthetic management.

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